a Endpoint MAGeCK LFC and abundance

	Guide 1			Guide 2			Guide 3			Guide 4		
	LFC	sgCTRL	LOY	LFC	sgCTRL	LOY	LFC	sgCTRL	LOY	LFC	sgCTRL	LOY
DDX3Y	0.12	10.9	11.7	0.39	25.6	33.1	1.36	3.2	8.0	-0.47	15.8	11.2
EIF1AY	-0.49	26.7	18.6	-0.76	21.0	12.2	0.52	5.0	7.0	0.52	2.6	3.6
RPS4Y1	-0.49	17.5	12.3	-1.17	2.2	1.0	0.07	7.0	7.2	0.45	14.2	19.2
RPS4Y2	0.86	11.1	19.9	0.30	27.4	33.2	0.65	9.6	14.9	0.21	18.2	20.6
ZFY	0.24	9.3	10.7	0.91	7.1	13.1	0.19	10.8	12.1	0.58	12.6	18.5
KDM5D	0.09	11.4	11.9	0.07	14.9	15.3	0.26	13.8	16.3	-1.03	4.1	2.0
UTY	0.75	13.4	22.2	-0.24	26.7	22.1	-0.27	23.9	19.4	0.08	31.3	32.6
USP9Y	0.61	16.7	25.1	0.14	23.4	25.4	-0.12	17.7	15.9	-0.81	46.2	25.8
AMELY	0.12	31.9	34.1	0.52	15.9	22.4	0.48	19.6	26.8	-0.07	12.5	11.7
BPY2	0.60	32.4	48.1	0.07	13.6	14.0	0.35	26.5	33.3	-0.41	10.2	7.5
BPY2B	0.95	7.9	15.0	-0.14	33.2	29.7	1.05	12.1	24.7	1.18	7.3	16.3
BPY2C	0.66	10.6	16.5	-0.53	10.3	7.0	-0.59	7.5	4.9	0.31	23.7	28.9
CDY1	0.83	12.8	22.4	0.39	15.3	19.7	0.24	41.0	47.5	0.28	21.7	25.9
CDY1B	0.42	11.2	14.7	0.29	38.6	46.4	0.50	11.4	15.9	0.68	27.6	43.6
CDY2A	0.99	8.7	17.0	0.52	10.2	14.4	0.15	12.1	13.2	0.00	26.6	26.1
CDY2B	0.20	21.6	24.3	0.11	31.2	33.2	0.20	8.0	9.1	-0.08	9.2	8.6
DAZ1	-0.08	9.8	9.1	-0.03	21.7	21.0	1.10	5.2	11.0	-0.69	10.8	6.6
DAZ2	0.65	3.3	5.1	0.40	1.4	1.9	0.26	13.6	16.0	0.25	8.6	10.0
DAZ3	0.84	5.2	9.1	0.59	5.3	7.9	0.79	21.2	36.0	1.55	5.5	15.7
DAZ4	0.49	15.9	21.9	0.83	12.1	21.1	2.26	1.1	5.2	1.43	4.0	10.6
HSFY1	0.20	36.6	41.3	0.94	5.6	10.5	-0.05	18.3	17.4	0.07	13.9	14.3
HSFY2	0.52	10.5	14.7	0.05	24.3	24.7	-0.06	14.9	14.0	0.36	7.2	9.1
NLGN4Y	0.10	22.2	23.3	-0.39	19.2	14.4	0.07	24.3	25.1	0.41	34.0	44.3
PCDH11Y	-0.26	21.0	17.3	0.38	22.1	28.2	-0.11	39.4	35.9	-0.28	19.9	16.1
PRORY	1.00	14.1	27.7	0.12	21.3	22.7	0.60	5.4	8.0	-0.53	18.1	12.3
PRY	0.78	12.0	20.3	*	0*	0*	0.09	15.5	16.2	1.37	1.4	3.6
PRY2	-0.02	20.6	20.1	-0.03	14.1	13.5	-0.28	15.0	12.1	-0.38	15.3	11.5
RBMY1A1	1.80	4.9	16.7	1.20	22.0	49.6	-0.11	5.7	5.2	1.14	6.5	14.0
RBMY1B	0.87	3.1	5.6	1.35	1.9	4.8	0.66	5.0	7.8	0.34	5.5	6.9
RBMY1D	-0.11	8.6	7.8	1.96	4.8	18.5	1.38	10.9	28.0	0.30	4.1	4.9
RBMY1E	1.99	4.0	15.7	0.37	8.8	11.2	0.67	11.2	17.5	0.60	13.4	19.9
RBMY1F	0.70	5.4	8.7	0.77	12.2	20.4	0.34	4.3	5.3	-2.38	0.7	0.1
RBMY1J	1.25	6.0	13.9	0.66	7.2	11.2	0.97	10.1	19.3	-0.41	4.7	3.5
SRY	-0.22	8.9	7.5	-0.20	21.4	18.3	0.22	12.6	14.4	0.67	7.5	11.7
TBL1Y	0.81	1.2	2.0	0.24	17.5	20.2	1.33	7.5	18.4	-0.39	11.8	8.9
TGIF2LY	-0.04	21.5	20.6	0.99	6.6	12.8	-0.06	17.2	16.2	-0.02	33.6	32.5
TMSB4Y	0.45	12.2	16.4	0.74	39.7	65.2	0.27	25.3	30.0	1.21	2.3	5.2
TSPY1	1.94	4.5	17.0	0.37	15.7	19.9	1.42	0.7	1.9	1.09	5.5	11.6
TSPY10	0.94	13.1	24.7	1.14	10.5	22.7	1.00	12.4	24.4	1.04	6.9	13.9
TSPY2	1.48	4.7	13.0	0.71	17.2	27.6	0.88	13.9	25.2	1.21	3.1	7.0
TSPY3	1.05	11.7	23.7	0.93	13.0	24.4	0.54	0.8	1.1	0.52	9.4	13.3
TSPY4	1.61	6.5	19.5	1.27	9.7	23.0	0.65	7.5	11.6	0.80	8.7	14.9
TSPY8	0.75	7.1	11.7	2.97	0.2	1.4	0.37	5.4	6.8	1.82	5.9	20.3
VCY	0.09	17.5	18.3	-0.44	45.4	33.0	1.18	0.5	1.2	0.04	47.0	47.3
VCY1B	0.68	12.3	19.2	0.62	0.6	0.9	0.38	18.2	23.2	-0.47	26.3	18.6
XKRY	-0.20	23.4	20.0	1.82	1.7	5.9	*	0*	0*	0.01	49.6	49.2
XKRY2	-0.26	11.5	9.4	0.06	11.1	11.3	-0.99	15.7	7.7	-0.95	3.7	1.9

b D0 LFC and abundance · one sample per genotype

D0 LFC = $\log_2[(\text{LOY CPM} + 1)/(\text{sgCTRL CPM} + 1)]$

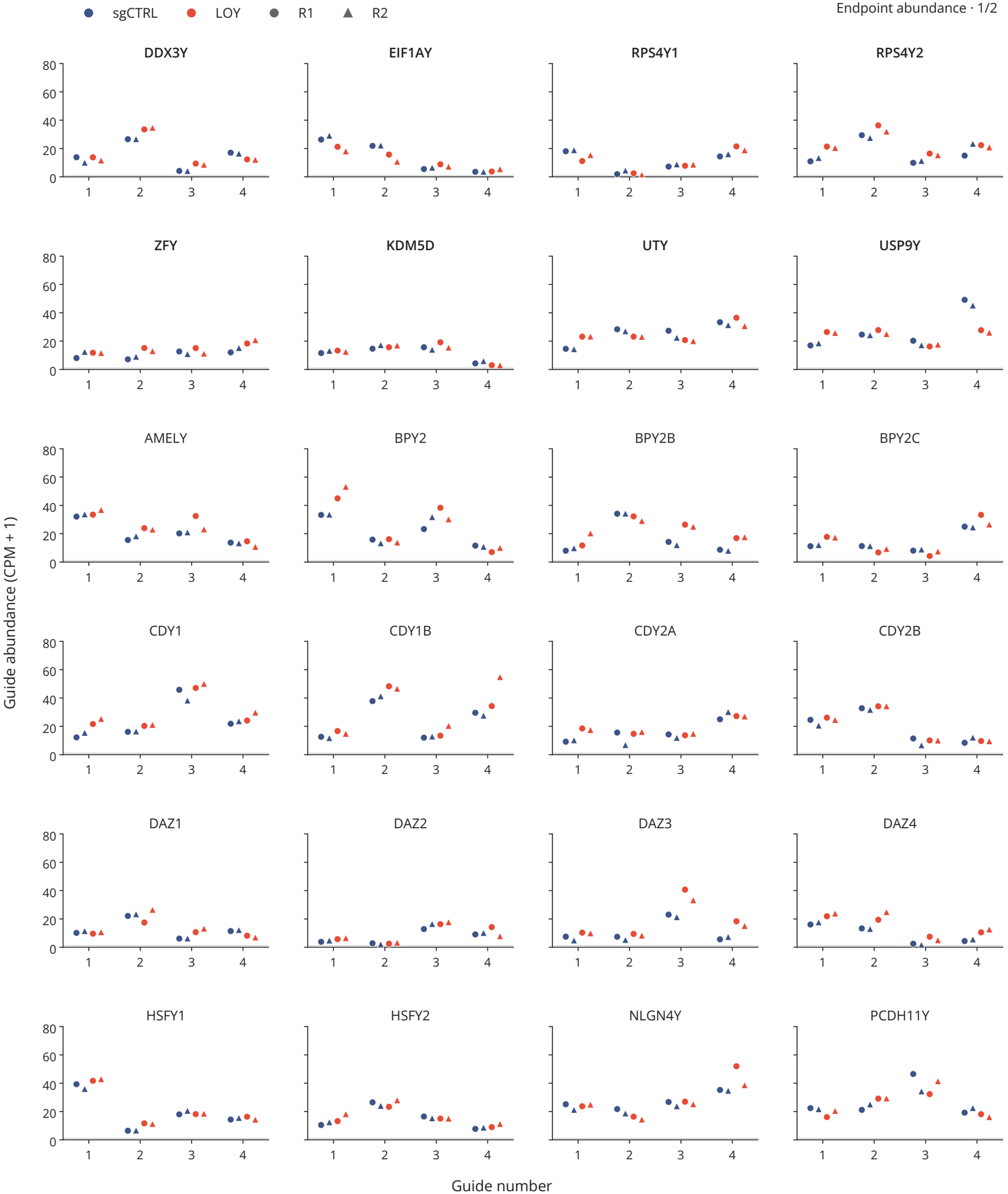
	Guide 1			Guide 2			Guide 3			Guide 4		
	LFC	sgCTRL	LOY	LFC	sgCTRL	LOY	LFC	sgCTRL	LOY	LFC	sgCTRL	LOY
DDX3Y	-0.24	2.18	1.70	0.47	2.31	3.58	0.83	0.50	1.67	0.31	1.50	2.11
EIF1AY	0.10	2.82	3.09	-0.18	2.31	1.91	0.60	1.14	2.24	-0.07	0.63	0.55
RPS4Y1	-0.25	2.18	1.67	0.01	0.21	0.22	0.36	0.68	1.15	0.19	1.77	2.16
RPS4Y2	0.23	1.63	2.08	0.08	3.91	4.18	0.44	1.16	1.94	0.15	1.75	2.05
ZFY	0.16	1.23	1.50	0.27	1.22	1.67	0.07	1.59	1.72	-0.10	1.77	1.59
KDM5D	0.06	1.93	2.05	0.02	1.25	1.29	0.35	1.57	2.27	0.16	0.34	0.49
UTY	0.65	2.23	4.07	-0.03	2.72	2.65	0.26	2.61	3.34	-0.45	4.09	2.73
USP9Y	-0.29	2.63	1.97	0.12	2.77	3.09	-0.08	2.36	2.19	-0.07	4.04	3.80
AMELY	0.14	3.09	3.50	0.15	1.77	2.08	0.14	2.63	3.01	0.18	2.52	2.98
BPY2	0.49	4.17	6.23	0.06	1.50	1.61	0.24	3.99	4.89	-0.16	0.91	0.71
BPY2B	0.50	1.82	2.98	0.26	3.86	4.84	0.11	2.07	2.32	0.24	1.61	2.08
BPY2C	0.32	0.89	1.37	0.11	0.97	1.12	0.03	0.89	0.93	0.32	2.63	3.53
CDY1	0.40	1.79	2.68	0.19	2.27	2.73	0.05	5.45	5.69	0.06	2.57	2.73
CDY1B	0.33	0.95	1.45	0.15	4.18	4.76	0.42	1.18	1.91	0.08	2.90	3.12
CDY2A	0.75	0.77	1.97	0.24	1.16	1.56	-0.14	2.50	2.19	0.20	3.58	4.27
CDY2B	0.25	2.93	3.69	0.35	3.72	5.03	0.34	1.20	1.78	0.24	0.93	1.29
DAZ1	0.51	1.00	1.86	-0.06	2.56	2.41	0.36	0.79	1.29	0.02	1.20	1.23
DAZ2	0.29	0.63	0.98	0.07	0.27	0.33	0.42	1.82	2.79	0.21	0.95	1.26
DAZ3	-0.17	1.05	0.82	-0.25	1.00	0.68	0.10	2.43	2.68	0.34	1.25	1.86
DAZ4	0.02	1.73	1.78	0.24	1.61	2.08	0.28	0.43	0.74	0.46	0.66	1.29
HSFY1	0.65	3.68	6.37	0.23	0.88	1.20	-0.10	2.32	2.11	0.16	1.48	1.78
HSFY2	0.34	1.09	1.64	0.40	2.59	3.75	0.45	1.05	1.80	-0.01	0.89	0.88
NLGN4Y	0.03	3.06	3.14	0.00	1.89	1.89	-0.13	3.07	2.73	0.16	3.84	4.40
PCDH11Y	0.21	2.36	2.90	0.14	3.70	4.18	0.04	5.01	5.17	0.18	1.82	2.19
PRORY	0.57	2.00	3.45	-0.09	3.09	2.84	0.16	1.02	1.26	-0.20	2.47	2.02
PRY	0.10	1.95	2.16	-0.08	0.054	0*	-0.03	2.15	2.08	0.04	0.32	0.36
PRY2	0.26	2.29	2.93	0.48	1.43	2.38	-0.16	1.81	1.50	-0.25	2.02	1.53
RBMY1A1	0.66	1.18	2.43	0.56	4.97	7.82	0.16	0.80	1.01	0.34	0.98	1.50
RBMY1B	0.15	0.64	0.82	0.31	0.46	0.82	0.69	0.52	1.45	0.16	0.73	0.93
RBMY1D	-0.19	0.95	0.71	0.86	0.86	2.38	0.22	2.22	2.73	0.09	0.55	0.66
RBMY1E	0.46	1.00	1.75	0.51	1.39	2.41	0.63	1.57	2.98	0.39	2.25	3.25
RBMY1F	0.33	0.95	1.45	0.44	1.47	2.35	0.02	0.98	1.01	0.01	0.072	0.082
RBMY1J	0.57	1.16	2.22	0.65	0.77	1.78	0.57	1.00	1.97	0.16	0.61	0.79
SRY	-0.05	1.29	1.20	0.14	2.18	2.52	0.17	2.04	2.41	0.27	1.13	1.56
TBL1Y	0.47	0.23	0.71	0.22	2.45	3.01	0.23	7.19	8.59	0.11	1.20	1.37
TGIF2LY	0.27	2.34	3.04	0.50	1.38	2.35	-0.33	2.36	1.67	-0.05	4.43	4.27
TMSB4Y	0.39	1.41	2.16	0.16	4.81	5.50	0.06	3.65	3.86	0.32	0.30	0.63
TSPY1	0.25	1.02	1.39	-0.11	2.38	2.13	0.04	0.27	0.30	0.37	1.14	1.78
TSPY10	0.89	1.79	4.18	0.51	1.54	2.63	0.86	1.64	3.80	0.36	1.27	1.91
TSPY2	0.50	0.91	1.70	0.44	2.47	3.69	0.58	1.98	3.47	0.70	0.54	1.50
TSPY3	0.47	2.15	3.36	0.54	2.82	4.57	0.02	0.18	0.19	0.08	1.45	1.59
TSPY4	0.75	1.54	3.28	0.81	1.52	3.42	0.51	1.23	2.19	0.32	1.43	2.02
TSPY8	0.46	1.14	1.94	0.01	0.018	0.027	0.09	1.50	1.67	0.49	1.70	2.79
VCY	-0.06	4.56	4.35	0.06	4.77	5.03	-0.02	0.13	0.11	0.02	6.10	6.21
VCY1B	0.15	1.72	2.02	0.05	0.36	0.41	0.44	2.04	3.12	0.26	2.54	3.23
XKRY	0.38	3.15	4.40	0.51	0.36	0.93	*	0*	0*	0.39	4.92	6.73
XKRY2	-0.25	0.88	0.57	-0.14	1.66	1.42	-0.11	1.77	1.56	0.47	0.38	0.90

c Gene-level drugZ

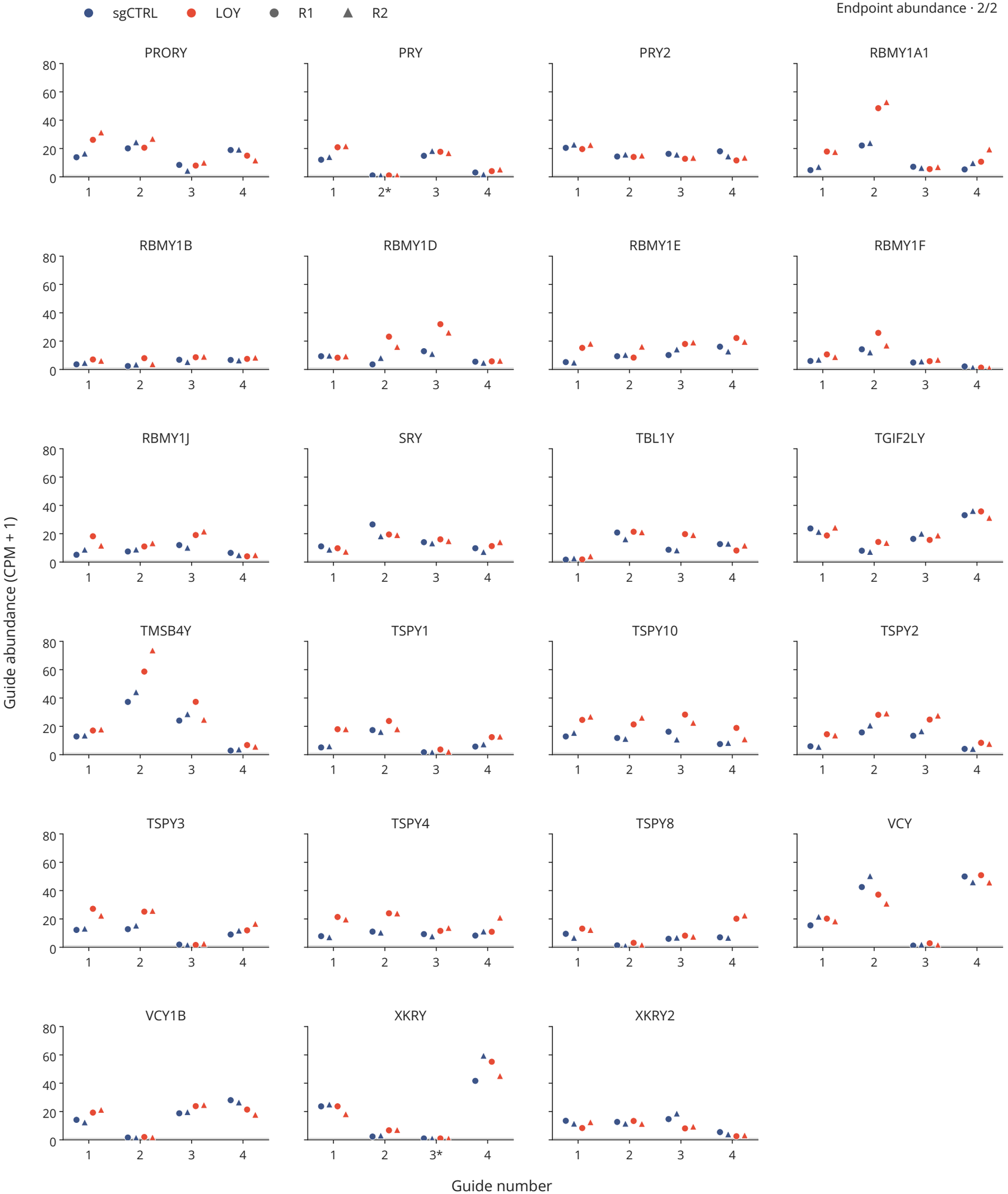
	Depletion FDR	Enrichment FDR
DDX3Y	0.986	0.980
EIF1AY	0.986	0.984
RPS4Y1	0.986	0.980
RPS4Y2	0.996	0.980
ZFY	0.994	0.980
KDM5D	0.986	0.980
UTY	0.986	0.980
USP9Y	0.986	0.981
AMELY	0.986	0.980
BPY2	0.986	0.980
BPY2B	0.999	0.980
BPY2C	0.986	0.980
CDY1	0.995	0.980
CDY1B	0.997	0.980
CDY2A	0.992	0.980
CDY2B	0.986	0.980
DAZ1	0.986	0.980
DAZ2	0.986	0.980
DAZ3	1.000	0.919
DAZ4	1.000	0.668
HSFY1	0.986	0.980
HSFY2	0.986	0.980
NLGN4Y	0.986	0.980
PCDH11Y	0.986	0.980
PRORY	0.986	0.980
PRY	0.995	0.980
PRY2	0.986	0.981
RBMY1A1	1.000	0.555
RBMY1B	0.996	0.980
RBMY1D	1.000	0.980
RBMY1E	1.000	0.980
RBMY1F	0.986	0.980
RBMY1J	0.996	0.980
SRY	0.986	0.980
TBL1Y	0.990	0.980
TGIF2LY	0.986	0.980
TMSB4Y	0.998	0.980
TSPY1	1.000	0.890
TSPY10	1.000	0.527
TSPY2	1.000	0.614
TSPY3	0.999	0.980
TSPY4	1.000	0.555
TSPY8	1.000	0.876
VCY	0.986	0.980
VCY1B	0.986	0.980
XKRY	0.986	0.980
XKRY2	0.986	0.996



0*: zero reads in the D0 sample or both endpoint cultures of that genotype. LFC *: absent from both genotypes at that timepoint.
Endpoint LFC: saved MAGeCK; D0 LFC: descriptive CPM + 1 contrast. Abundance colour scales differ. Original gene-level FDRs retained; no D0 adjustment.



* No endpoint reads. R1/R2: outgrowth cultures from one established population per genotype.



* No endpoint reads. R1/R2: outgrowth cultures from one established population per genotype.